CI/CD Pipeline Documentation

1.Aim

Configure a complete Continuous Integration and Continuous Deployment (CI/CD) pipeline using:

- Red Hat Enterprise Linux 9.8
- Jenkins
- Apache Maven
- Git & GitHub
- Apache Tomcat

The pipeline automatically builds a Java Maven Web Application from GitHub and deploys it to the Apache Tomcat server.

2. Prerequisites

Before starting, ensure:

- Two RHEL 9.8 Virtual Machines
  - Jenkins Server
  - Tomcat Server
- Internet connectivity
- Root privileges
- Mounted RHEL DVD repository
- Firewall service enabled

3. Jenkins Server Configuration

Step 1- Verify Network Connectivity

Configure the virtual machine to use the NAT network adapter and verify internet connectivity using the following commands:

```
#ping www.google.com
#ping www.redhat.com
```

```
[root@jenkins ~]#
[root@jenkins ~]# ping www.google.com
PING www.google.com (142.251.153.119) 56(84) bytes of data.
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=1 ttl=128 time=30.4 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=2 ttl=128 time=28.2 ms
^C
--- www.google.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 28.157/29.298/30.440/1.141 ms
[root@jenkins ~]# ping www.redhat.com
PING e3396.dscx.akamaiedge.net (23.63.221.112) 56(84) bytes of data.
64 bytes from a23-63-221-112.deploy.static.akamaitechnologies.com (23.63.221.112): icmp_seq=1 t
tl=128 time=33.6 ms
64 bytes from a23-63-221-112.deploy.static.akamaitechnologies.com (23.63.221.112): icmp_seq=2 t
tl=128 time=43.6 ms
^C
--- e3396.dscx.akamaiedge.net ping statistics ---
3 packets transmitted, 2 received, 33.3333% packet loss, time 2004ms
rtt min/avg/max/mdev = 33.638/38.631/43.624/4.993 ms
```

Step 2- Take VM Snapshot

Step 3- Configure Hostname

#hostnamectl set-hostname jenkins.example.com

```
[root@jenkins ~]#
[root@jenkins ~]# hostnamectl set-hostname jenkins.example.com
[root@jenkins ~]# hostname
jenkins.example.com
```

Step 4- Mount RHEL DVD

#mount /dev/cdrom /mnt
#yum clean all
#yum repolist all

```
[root@jenkins ~]# mount /dev/cdrom /mnt
mount: /mnt: WARNING: source write-protected, mounted read-only.
[root@jenkins ~]# yum clean all
Updating Subscription Management repositories.

This system is registered with an entitlement server, but is not receiving updates. You can use
 subscription-manager to assign subscriptions.

36 files removed
[root@jenkins ~]# yum repolist all
Updating Subscription Management repositories.

This system is registered with an entitlement server, but is not receiving updates. You can use
 subscription-manager to assign subscriptions.

repo id                                                       repo name                 status
AppStream                                                     this is cdrom repo of ap  enabled
BaseOS                                                        This is a baseos repo     enabled
```

Step 5- Install Java 21 and Verify installation

Java is required because both Jenkins and Maven run on Java. Install Java using the following command:

#yum install java-21-openjdk -y

#java –version

```
Complete!
[root@jenkins ~]# java --version
openjdk 21.0.11 2026-04-21 LTS
OpenJDK Runtime Environment (Red_Hat-21.0.11.0.10-1) (build 21.0.11+10-LTS)
OpenJDK 64-Bit Server VM (Red_Hat-21.0.11.0.10-1) (build 21.0.11+10-LTS, mixed mode, sharing)
```

Step 6- Apache Maven Installation

Download Maven

#cd /opt
#wget https://dlcdn.apache.org/maven/maven-3/3.9.16/binaries/apache-maven-3.9.16-bin.tar.gz

```
[root@jenkins ~]# cd /opt
[root@jenkins opt]# wget https://dlcdn.apache.org/maven/maven-3/3.9.16/binaries/apache-maven-3.9.
16-bin.tar.gz
--2026-07-02 22:26:41--  https://dlcdn.apache.org/maven/maven-3/3.9.16/binaries/apache-maven-3.9.
16-bin.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9278065 (8.8M) [application/x-gzip]
Saving to: ‘apache-maven-3.9.16-bin.tar.gz’

apache-maven-3.9.16-bin. 100%[===============================>]   8.85M   362KB/s    in 13s

2026-07-02 22:26:57 (709 KB/s) - ‘apache-maven-3.9.16-bin.tar.gz’ saved [9278065/9278065]
```

Step 7- Extract and rename

#tar -xvzf apache-maven-3.9.16-bin.tar.gz
#mv apache-maven-3.9.16 maven

```
[root@jenkins opt]# tar -xvzf apache-maven-3.9.16-bin.tar.gz
apache-maven-3.9.16/README.txt
apache-maven-3.9.16/LICENSE
apache-maven-3.9.16/NOTICE
apache-maven-3.9.16/lib/
apache-maven-3.9.16/lib/aopalliance.license
apache-maven-3.9.16/lib/asm.license
apache-maven-3.9.16/lib/commons-cli.license
```

Step 8- Configure Environment Variables

Edit ~/.bashrc

Add the following environment variables to configure Java and Maven paths-

JAVA_HOME=/usr/lib/jvm/java-21-openjdk
M2_HOME=/opt/maven
M2=/opt/maven/bin
PATH=$PATH:$HOME/bin:$JAVA_HOME:$M2:$M2_HOME

export PATH

```
alias rm='rm -i'
alias cp='cp -i'
alias mv='mv -i'

########for configure path of maven#######
JAVA_HOME=/usr/lib/jvm/java-21-openjdk

M2_HOME=/opt/maven

M2=/opt/maven/bin

PATH=$PATH:$HOME/bin:$JAVA_HOME:$M2:$M2_HOME

export PATH

-- INSERT --
```

Reload the .bashrc file so the newly added environment variables take effect.

source ~/.bashrc

```
[root@jenkins ~]# source .bashrc
[root@jenkins ~]# mvn -version
Apache Maven 3.9.16 (2bdd9fddda4b155ebf8000e807eb73fd829a51d5)
Maven home: /opt/maven
Java version: 17.0.19, vendor: Red Hat, Inc., runtime: /usr/lib/jvm/java-17-openjdk-17.0.19.0.10-2.el9.x86_64
Default locale: en_US, platform encoding: UTF-8
OS name: "linux", version: "5.14.0-687.17.1.el9_8.x86_64", arch: "amd64", family: "unix"
[root@jenkins ~]#
```

Step 9- Create Maven Web Application

mvn archetype:generate | grep maven-archetype-webapp

mvn archetype:generate

```
[root@jenkins ~]# mvn archetype:generate | grep maven-archetype-webapp
740: remote -> com.haoxuer.maven.archetype:maven-archetype-webapp (a simple maven archetype)
903: remote -> com.lodsve:lodsve-maven-archetype-webapp (Lodsve Maven Archetype Webapp)
2362: remote -> org.apache.maven.archetypes:maven-archetype-webapp (An archetype which contains a sample Maven Webapp project.)
2630: remote -> org.bytesizebook.com.guide.boot:maven-archetype-webapp (An archetype to create the starting code for the first three chapters of Guide to Web Development with Java, 2nd edition.)
^C
[root@jenkins ~]# mvn archetype:generate
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Interactive mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
Choose archetype:
```

```
3681: remote -> za.co.absa.hyperdrive:component-archetype (-)
3682: remote -> za.co.absa.hyperdrive:component-archetype_2.11 (-)
3683: remote -> za.co.absa.hyperdrive:component-archetype_2.12 (-)
Choose a number or apply filter (format: [groupId:]artifactId, case sensitive contains): 2357: 2362
Choose org.apache.maven.archetypes:maven-archetype-webapp version:
1: 1.0-alpha-1
2: 1.0-alpha-2
3: 1.0-alpha-3
4: 1.0-alpha-4
5: 1.0
6: 1.3
7: 1.4
8: 1.5
Choose a number: 8: 8
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webap
p/1.5/maven-archetype-webapp-1.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp
/1.5/maven-archetype-webapp-1.5.jar (9.9 kB at 9.9 kB/s)
Define value for property 'groupId': in.radical.webapp
Define value for property 'artifactId': radical-webapp
Define value for property 'version' 1.0-SNAPSHOT:
Define value for property 'package' in.radical.webapp:
Confirm properties configuration:
groupId: in.radical.webapp
artifactId: radical-webapp
version: 1.0-SNAPSHOT
package: in.radical.webapp
 Y: Y
```

<u>The Maven Archetype generates the initial source code and project structure for the web application. This generated project will be used throughout the CI/CD pipeline. Navigate to the generated project directory (radical-webapp) and continue with the remaining steps.</u>

After the project is created, verify that the directory exists. Run:

#ls

Navigate into the project directory.

#cd radical-webapp

Display the project structure.

#tree .

```
[root@jenkins ~]# pwd
/root
[root@jenkins ~]# ls
adduser.yml      Desktop    Downloads  Pictures  radical-webapp  Videos
anaconda-ks.cfg  Documents  Music      Public    Templates
[root@jenkins ~]# cd radical-webapp/
[root@jenkins radical-webapp]# ll
total 4
-rw-r--r--. 1 root root 2220 Jul  2 22:52 pom.xml
drwxr-xr-x. 3 root root   18 Jul  2 22:52 src
[root@jenkins radical-webapp]# tree .
.
├── pom.xml
└── src
    └── main
        └── webapp
            ├── index.jsp
            └── WEB-INF
                └── web.xml

4 directories, 3 files
[root@jenkins radical-webapp]#
```

Run the Maven lifecycle commands one by one to verify that the project builds successfully.

- #mvn validate - Validates the project structure.
- #mvn compile - Compiles the source code.
- #mvn test - Executes unit tests.
- #mvn package - Packages the application and generates the WAR file.

```
cd radical-webapp/
ll
tree .
mvn validate
mvn test
mvn compile
mvn test
mvn package
tree .
history
```

```
[root@jenkins radical-webapp]# tree .
.
├── pom.xml
├── src
│   └── main
│       └── webapp
│           ├── index.jsp
│           └── WEB-INF
│               └── web.xml
└── target
    ├── maven-archiver
    │   └── pom.properties
    ├── radical-webapp
    │   ├── index.jsp
    │   ├── META-INF
    │   └── WEB-INF
    │       ├── classes
    │       └── web.xml
    └── radical-webapp.war

10 directories, 7 files
[root@jenkins radical-webapp]#
```

Step 10- Git & GitHub Configuration

Install Git

yum install git -y

```
Installed products updated.

Downgraded:
  git-core-2.39.1-1.el9.x86_64
Installed:
  git-2.39.1-1.el9.x86_64                         git-core-doc-2.39.1-1.el9.noarch
  perl-Error-1:0.17029-7.el9.noarch               perl-Git-2.39.1-1.el9.noarch
  perl-TermReadKey-2.38-11.el9.x86_64             perl-lib-0.65-480.el9.x86_64

Complete!
[root@jenkins ~]# git --version
git version 2.39.1
[root@jenkins ~]#
```

Initialize the git repository

#git init .

```
[root@jenkins radical-webapp]# git init .
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /root/radical-webapp/.git/
```

Configure Git by adding your username and email.

#git config --global user.name "andyy"
#git config --global user.email “andy@example.com”

```
root@jenkins:~/radical-webapp
[root@jenkins radical-webapp]# git config --global user.name "andyy"
[root@jenkins radical-webapp]#
[root@jenkins radical-webapp]# git config --global user.email "andy@example.com"
[root@jenkins radical-webapp]#
```

```
[root@jenkins radical-webapp]# git config --global --list
user.name=andyy
user.email=andy@example.com
[root@jenkins radical-webapp]#
```

Stage all files so they are ready to be committed.

git add .

```
[root@jenkins radical-webapp]# git add .
[root@jenkins radical-webapp]#
[root@jenkins radical-webapp]# git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .mvn/jvm.config
        new file:   .mvn/maven.config
        new file:   pom.xml
        new file:   src/main/webapp/WEB-INF/web.xml
        new file:   src/main/webapp/index.jsp
        new file:   target/maven-archiver/pom.properties
        new file:   target/radical-webapp.war
        new file:   target/radical-webapp/WEB-INF/web.xml
        new file:   target/radical-webapp/index.jsp
```

Once all required changes have been staged, create a commit to save the current version of the project.

git commit -m "Initial Commit" #git commit -m "code1 my originalcode"

```
[root@jenkins radical-webapp]# git commit -m "code1 my originalcode"
[master (root-commit) d68b7d3] code1 my originalcode
 9 files changed, 100 insertions(+)
 create mode 100644 .mvn/jvm.config
 create mode 100644 .mvn/maven.config
 create mode 100644 pom.xml
 create mode 100644 src/main/webapp/WEB-INF/web.xml
 create mode 100644 src/main/webapp/index.jsp
 create mode 100644 target/maven-archiver/pom.properties
 create mode 100644 target/radical-webapp.war
 create mode 100644 target/radical-webapp/WEB-INF/web.xml
 create mode 100644 target/radical-webapp/index.jsp
[root@jenkins radical-webapp]#
[root@jenkins radical-webapp]#
```

To verify we can see in git log

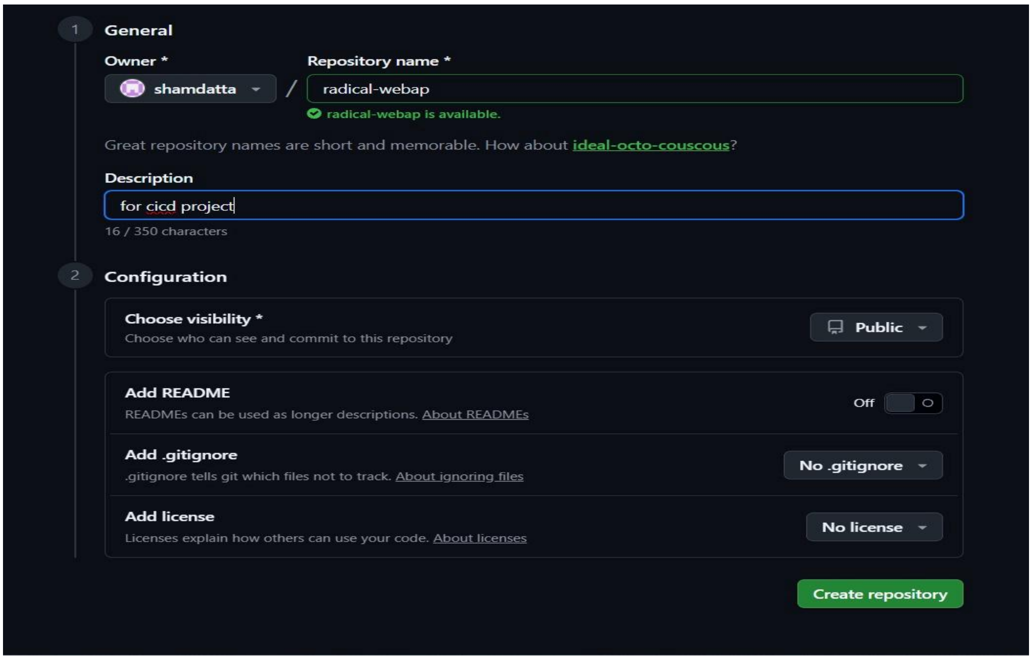
#git log

```
[root@jenkins radical-webapp]# git log
commit d68b7d32e9b312a16f456ce644f39234e11ac5a6 (HEAD -> master)
Author: andyy <andy@example.com>
Date:   Tue Jul 7 21:12:47 2026 +0530

    code1 my originalcode
[root@jenkins radical-webapp]#
```

Connect GitHub repository

Create a new repository in your GitHub account with the same name as your local project.



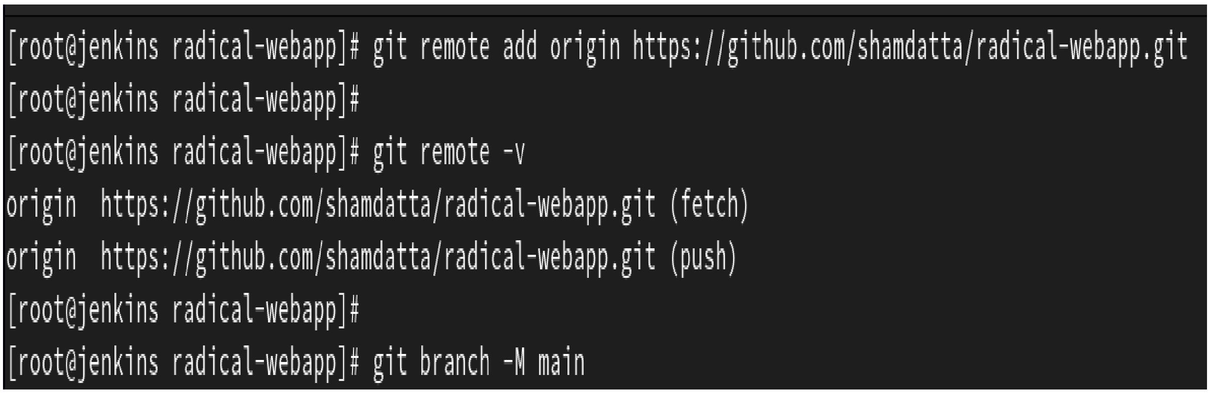
GITHUB PUSH

Connect the local Git repository to the GitHub repository and push the project.

#git remote add origin https://github.com/<username>/radical-webapp.git
#git branch -M main
#git push -u origin main

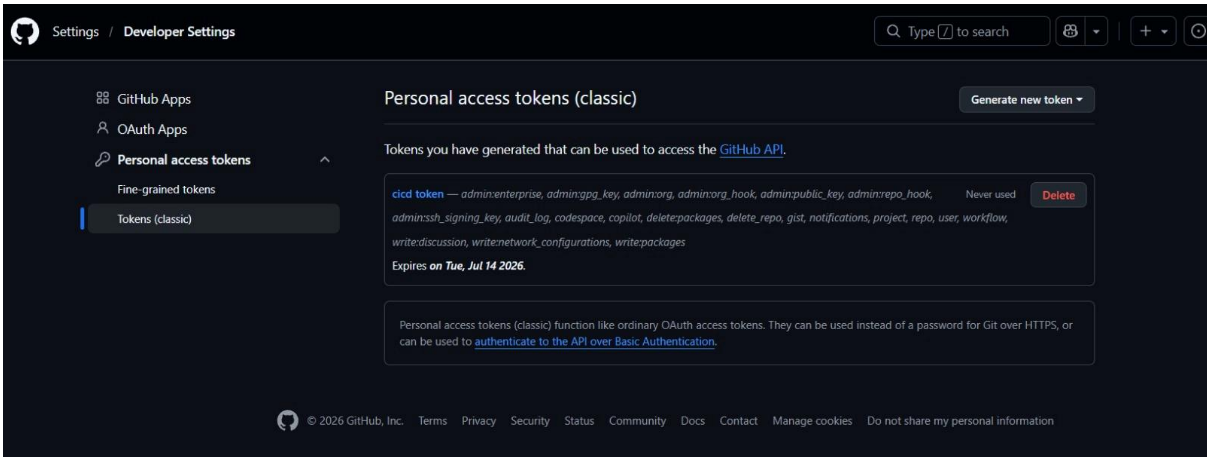
GitHub will prompt for authentication.

- Enter your GitHub username.
- For the password, use the Personal Access Token (Classic) generated instead of your GitHub account password.

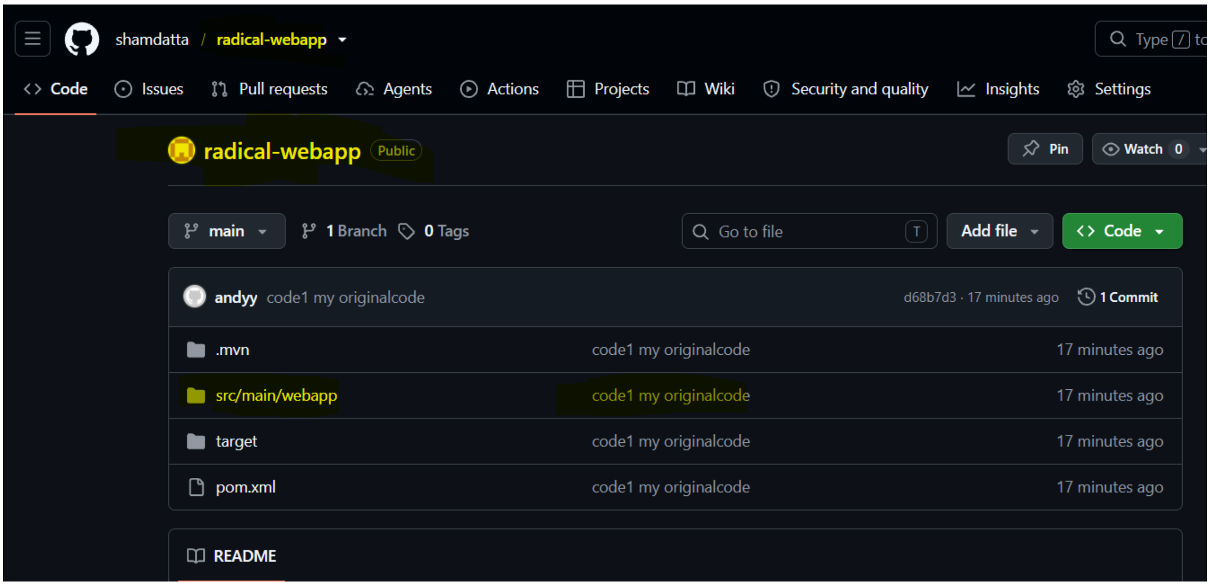


```
[root@jenkins radical-webapp]# git remote add origin https://github.com/shamdatta/radical-webapp.git
[root@jenkins radical-webapp]#
[root@jenkins radical-webapp]# git remote -v
origin  https://github.com/shamdatta/radical-webapp.git (fetch)
origin  https://github.com/shamdatta/radical-webapp.git (push)
[root@jenkins radical-webapp]#
[root@jenkins radical-webapp]# git branch -M main
```

```
[root@jenkins radical-webapp]# git push -u origin main
Username for 'https://github.com': andyy
Password for 'https://andyy@github.com':
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 2 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (15/15), 3.23 KiB | 1.62 MiB/s, done.
Total 15 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/shamdatta/radical-webapp.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
[root@jenkins radical-webapp]#
```



Verify that the project has been successfully pushed to GitHub by opening the repository in your browser.



Now make a small change to the project to verify that the changes can be pushed successfully.

#vi src/main/webapp/index.jsp

```
<html>
<body>
<h2><%= "Hello World! Welcome to our Project!!!"%></h2>
</body>
</html>
~
```

After making changes, stage the modified files using:

git add .

git status

```
[root@jenkins radical-webapp]# git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   src/main/webapp/index.jsp

no changes added to commit (use "git add" and/or "git commit -a")
[root@jenkins radical-webapp]# git add .
[root@jenkins radical-webapp]#
[root@jenkins radical-webapp]# git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   src/main/webapp/index.jsp
```

Once the modified file is in the staging area, commit the changes.

#git log

```
[root@jenkins radical-webapp]# git commit -m "code2 my 2nd code modification"
[main b9cb95b] code2 my 2nd code modification
 1 file changed, 1 insertion(+), 1 deletion(-)
[root@jenkins radical-webapp]#
[root@jenkins radical-webapp]# git log
commit b9cb95b9e5db27419863560e439cc6137994bcbd (HEAD -> main)
Author: andyy <andy@example.com>
Date:   Tue Jul 7 21:35:53 2026 +0530

    code2 my 2nd code modification

commit d68b7d32e9b312a16f456ce644f39234e11ac5a6 (origin/main)
Author: andyy <andy@example.com>
Date:   Tue Jul 7 21:12:47 2026 +0530

    code1 my originalcode
[root@jenkins radical-webapp]#
```

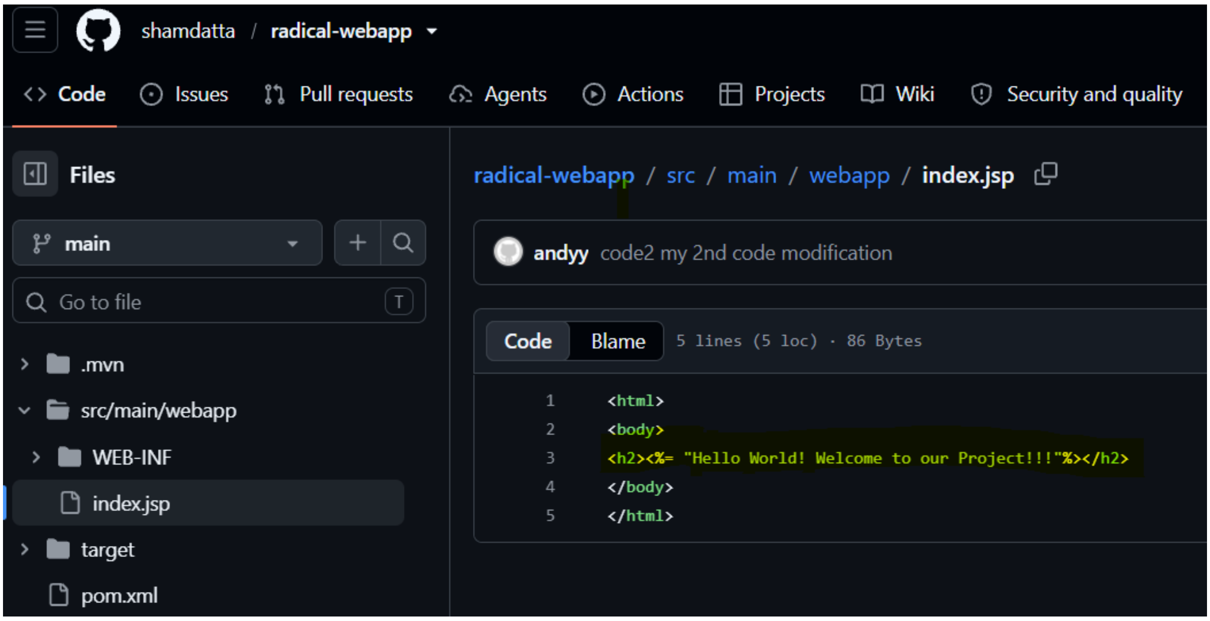
Push to GitHub

#git remote add origin https://github.com/username/radical-webapp
#git branch -M main
#git push -u origin main

- Add username and password (the classic token we generated)

```
[root@jenkins radical-webapp]# git push -u origin main
Username for 'https://github.com': andyy
Password for 'https://andyy@github.com':
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 495 bytes | 495.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/shamdatta/radical-webapp.git
   d68b7d3..b9cb95b  main -> main
branch 'main' set up to track 'origin/main'.
[root@jenkins radical-webapp]#
```

Verification -



Step 11- Install and configure Jenkins

#wget -O /etc/yum.repos.d/jenkins.repo \https://pkg.jenkins.io/rpm-stable/jenkins.repo

```
[root@jenkins radical-webapp]#  wget -O /etc/yum.repos.d/jenkins.repo \https://pkg.jenkins.io/rpm-stable/jenkins
.repo
--2026-07-07 21:43:49--  https://pkg.jenkins.io/rpm-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 151.101.210.133, 2a04:4e42:25::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|151.101.210.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.re 100%[=========================================>]     267  --.-KB/s    in 0s

2026-07-07 21:43:50 (14.9 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [267/267]

[root@jenkins radical-webapp]#
```

The Jenkins repository configuration file has been downloaded to /etc/yum.repos.d/.

This allows the yum package manager to install Jenkins from the official Jenkins repository.

Check using –

ll /etc/yum.repos.d

yum repolist

```
[root@jenkins yum.repos.d]# ll /etc/yum.repos.d/
total 332
-rw-r--r--. 1 root root    101 Jun 25 11:08 AppStream.repo
-rw-r--r--. 1 root root     85 Jun 25 11:08 BaseOS.repo
-rw-r--r--. 1 root root    267 Jun  9 12:44 jenkins.repo
-rw-r--r--. 1 root root 326945 Jul  7 15:17 redhat.repo
[root@jenkins yum.repos.d]#
[root@jenkins yum.repos.d]# yum repolist
Updating Subscription Management repositories.
repo id                                     repo name
AppStream                                   this is cdrom repo of appstream
BaseOS                                      This is a baseos repo
jenkins                                     Jenkins-stable
[root@jenkins yum.repos.d]#
```

Now we will install Jenkins

#yum install Jenkins

```
[root@jenkins yum.repos.d]#  yum install jenkins
Updating Subscription Management repositories.
Jenkins-stable                                                                  1.1 kB/s | 833  B     00:00
Jenkins-stable                                                                  1.6 kB/s | 1.6 kB     00:01
Importing GPG key 0x14ABFC68:
 Userid     : "Jenkins Project <jenkinsci-board@googlegroups.com>"
 Fingerprint: 5E38 6EAD B55F 0150 4CAE 8BCF 7198 F4B7 14AB FC68
 From       : https://pkg.jenkins.io/rpm-stable/repodata/repomd.xml.key
Is this ok [y/N]: y
Jenkins-stable                                                                  1.5 kB/s | 2.3 kB     00:01
Last metadata expiration check: 0:00:01 ago on Tue 07 Jul 2026 09:47:26 PM IST.
Dependencies resolved.
================================================================================================================
 Package                    Architecture              Version                     Repository                Size
================================================================================================================
Installing:
 jenkins                    noarch                    2.555.3-1                   jenkins                   95 M

Transaction Summary
================================================================================================================
Install  1 Package

Total download size: 95 M
Installed size: 95 M
Is this ok [y/N]: y
```

Purpose: Install Jenkins on the Linux server so it can automate software build, testing, and deployment tasks.

Enable Jenkins services

```
[root@jenkins yum.repos.d]# systemctl daemon-reload
[root@jenkins yum.repos.d]#
[root@jenkins yum.repos.d]# systemctl enable jenkins
[root@jenkins yum.repos.d]#
[root@jenkins yum.repos.d]# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
     Active: active (running) since Tue 2026-07-07 22:23:06 IST; 2min 28s ago
   Main PID: 11866 (java)
      Tasks: 37 (limit: 10324)
     Memory: 456.7M (peak: 463.4M)
        CPU: 29.123s
     CGroup: /system.slice/jenkins.service
             └─11866 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cac>

Jul 07 22:22:54 jenkins.example.com jenkins[11866]: [LF]> This may also be found at: /var/lib/jenkins/secrets/i>
Jul 07 22:22:54 jenkins.example.com jenkins[11866]: [LF]>
Jul 07 22:22:54 jenkins.example.com jenkins[11866]: [LF]> *****************************************************>
Jul 07 22:22:54 jenkins.example.com jenkins[11866]: [LF]> *****************************************************>
```

Configure Firewall

Allow TCP port 8080 through the firewall so Jenkins can be accessed from a web browser.

#firewall-cmd –permanent—add-port=8080/tcp

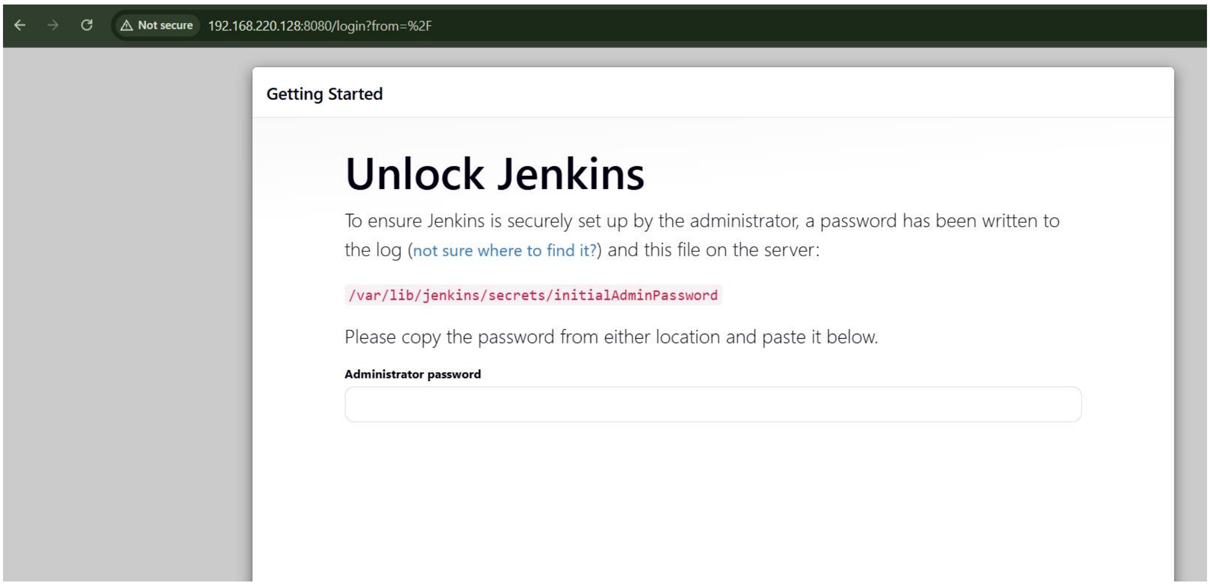
#firewall-cmd --reload

```
[root@jenkins yum.repos.d]# firewall-cmd --permanent --add-port=8080/tcp
success
[root@jenkins yum.repos.d]# firewall-cmd --reload
success
[root@jenkins yum.repos.d]#
```

Open the following URL in your web browser.

#http://192.168.220.126:8080

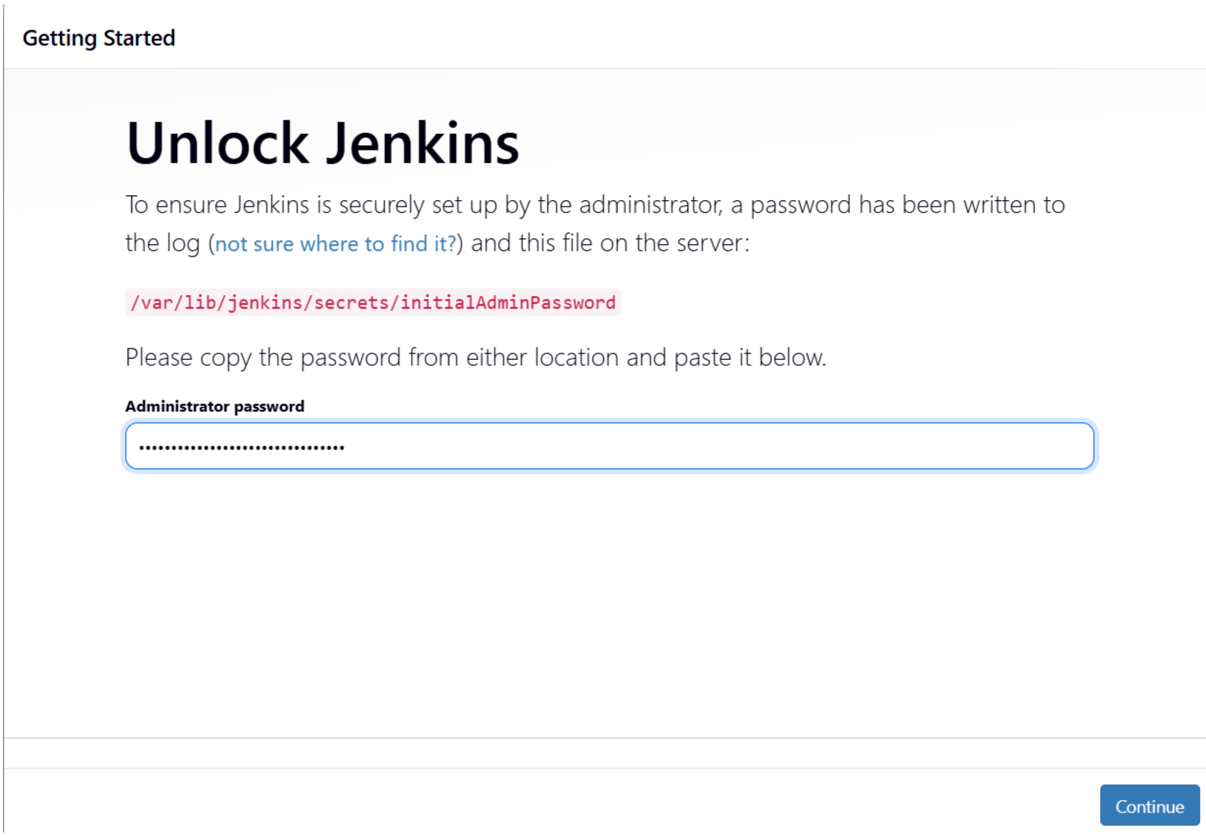
Verification- Jenkins is now accessible through port 8080.



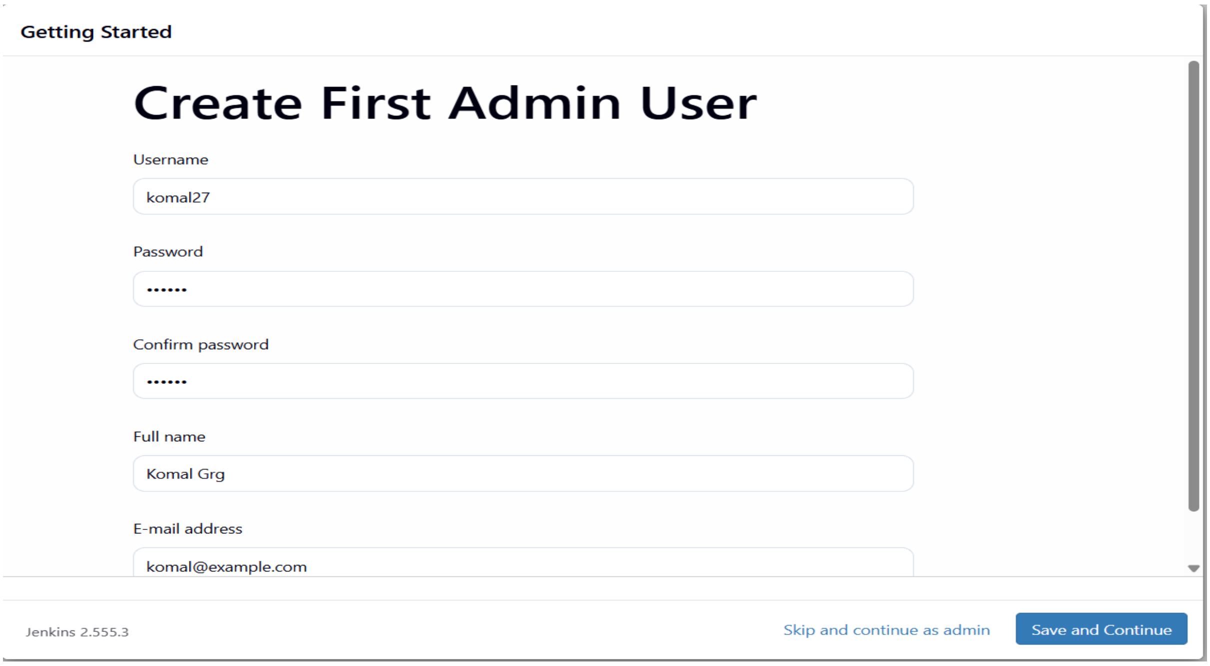
Retrieve the initial administrator password using the following command:

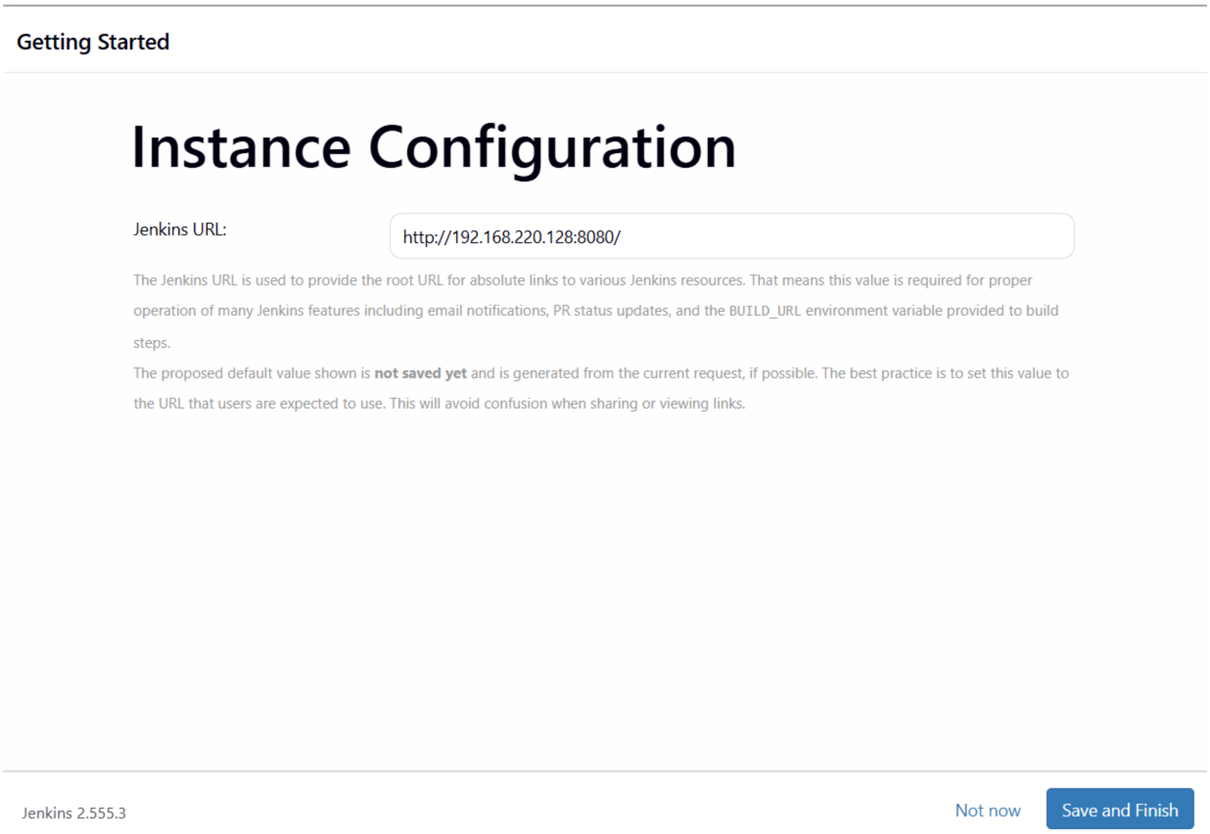
#cat /var/lib/jenkins/secrets/initialAdminPassword

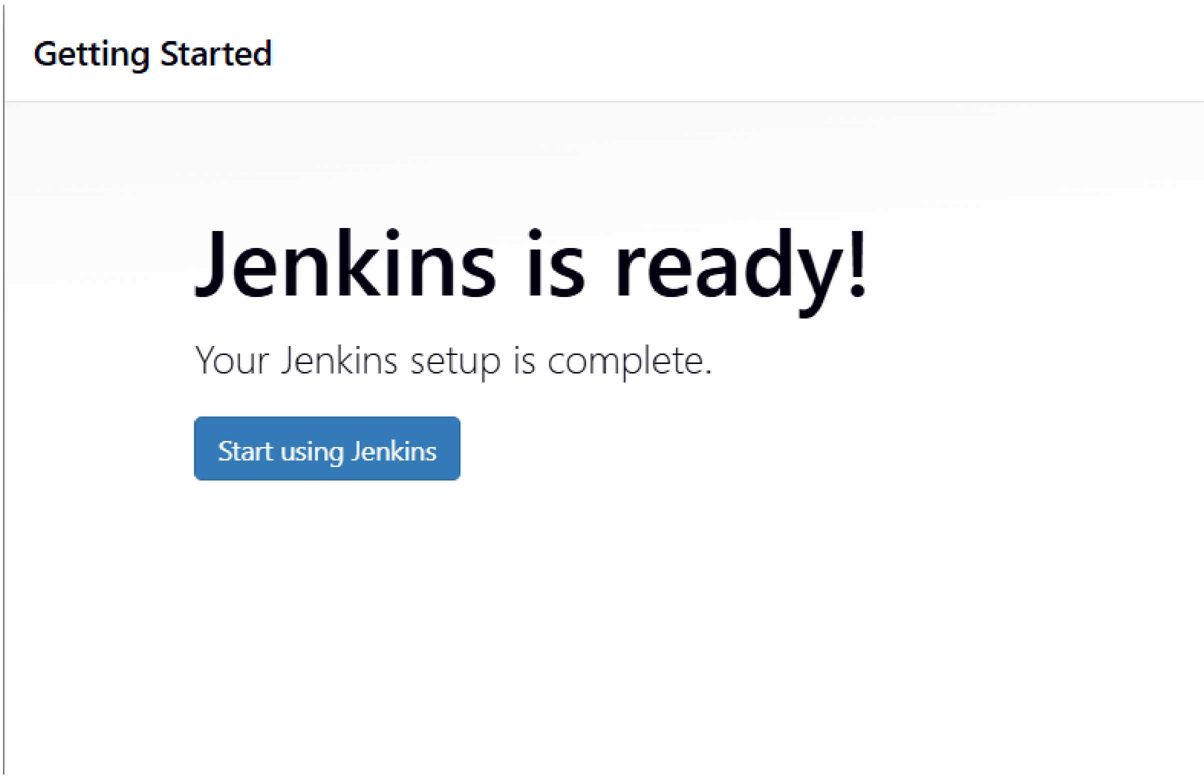
Now access Jenkins



Create the first administrator user for Jenkins.




Getting Started
Instance Configuration
Jenkins URL:
http://192.168.220.128:8080/
The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.
The proposed default value shown is not saved yet and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.
Jenkins 2.555.3
Not now
Save and Finish

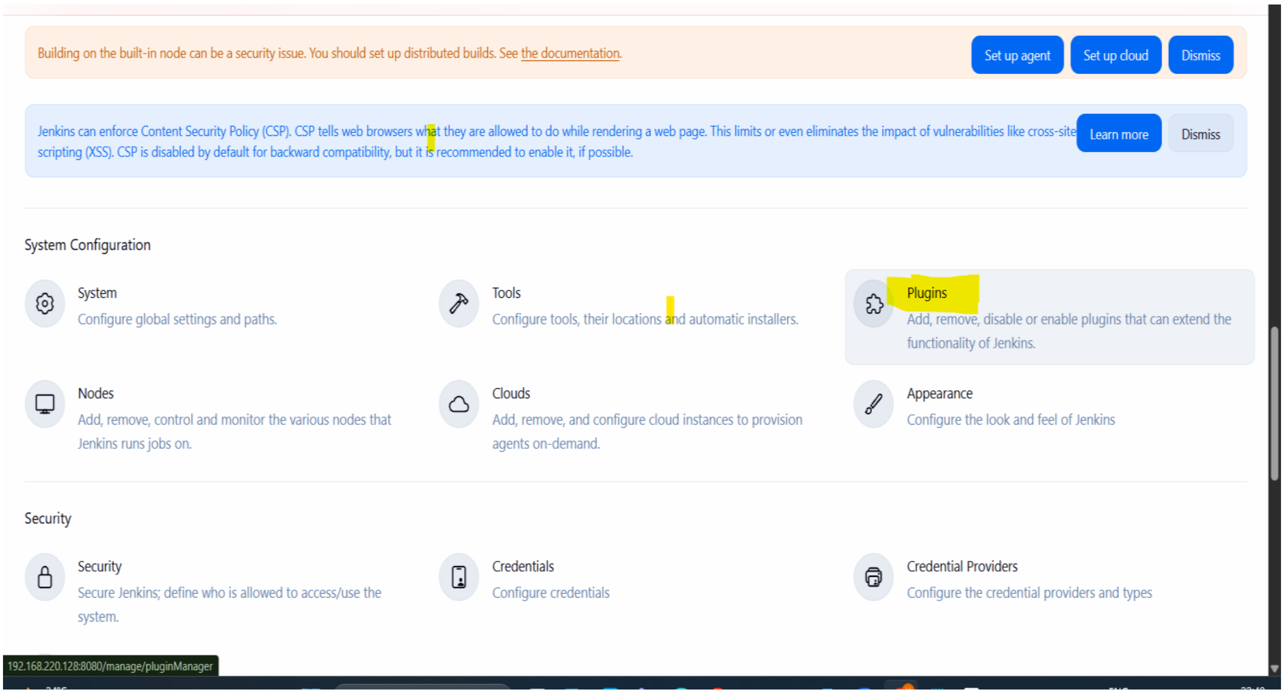

Getting Started
Jenkins is ready!
Your Jenkins setup is complete.
Start using Jenkins

The first administrator account has been created successfully. You can now access the Jenkins dashboard.

Navigate to:

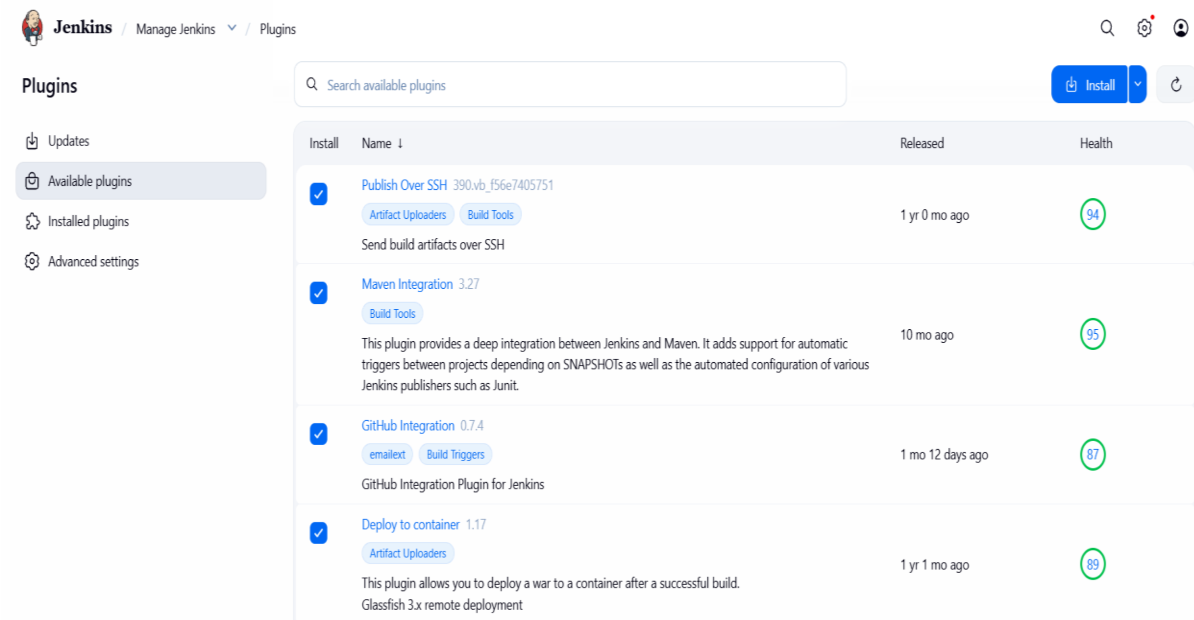
Dashboard→ Manage Jenkins→ Plugins

Install the following plugins required for the CI/CD pipeline.

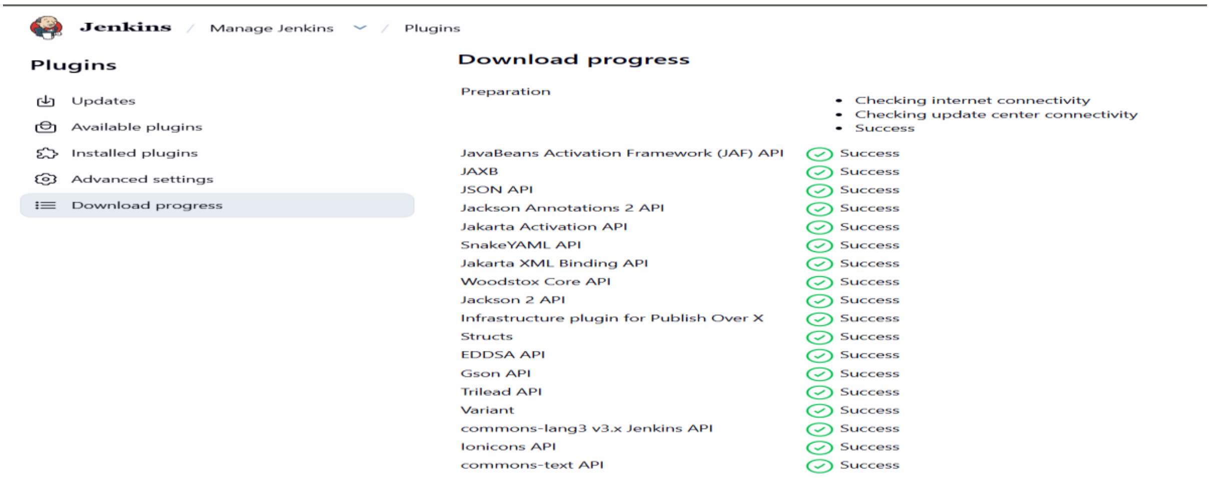


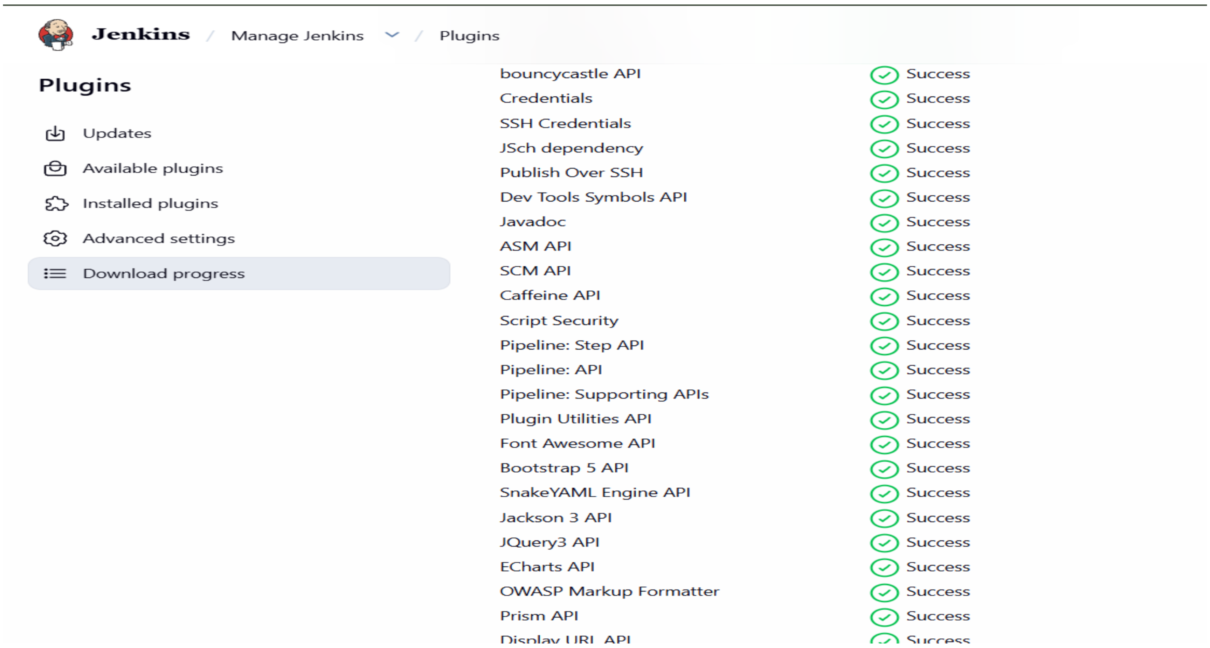
Install the following plugins required for the CI/CD pipeline:

- Publish over SSH
- Maven Integration
- GitHub Integration
- Deploy to container



Purpose: Plugins extend Jenkins functionality. In this document we install the plugins required for Pipeline, Git integration, and Maven builds.





All the required plugins have been installed successfully.

Run the following command to verify JAVA_HOME:

#echo $JAVA_HOME

```
[root@jenkins plugins]# echo $JAVA_HOME
/usr/lib/jvm/java-21-openjdk
[root@jenkins plugins]#
```

This path will be used while configuring the JDK in Jenkins.

Configure Global Tools

Navigate to:

Dashboard

→ Manage Jenkins

→ Tools

Configure the following tools:

• JDK

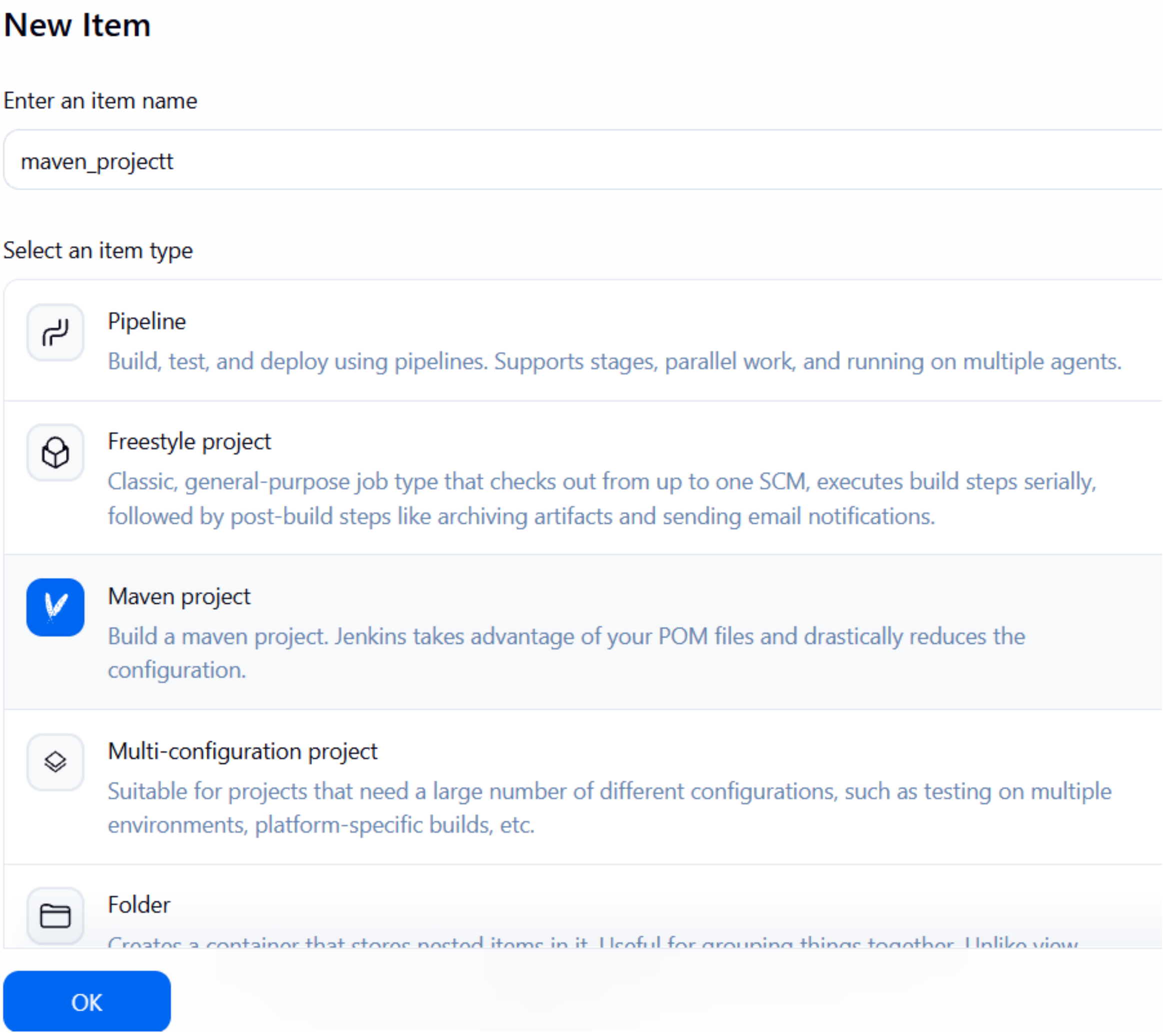
• Maven

These tools will be used by Jenkins while building the project.


Maven installations
Add Maven
Maven
Name
maven
MAVEN_HOME
/opt/maven
Install automatically
?

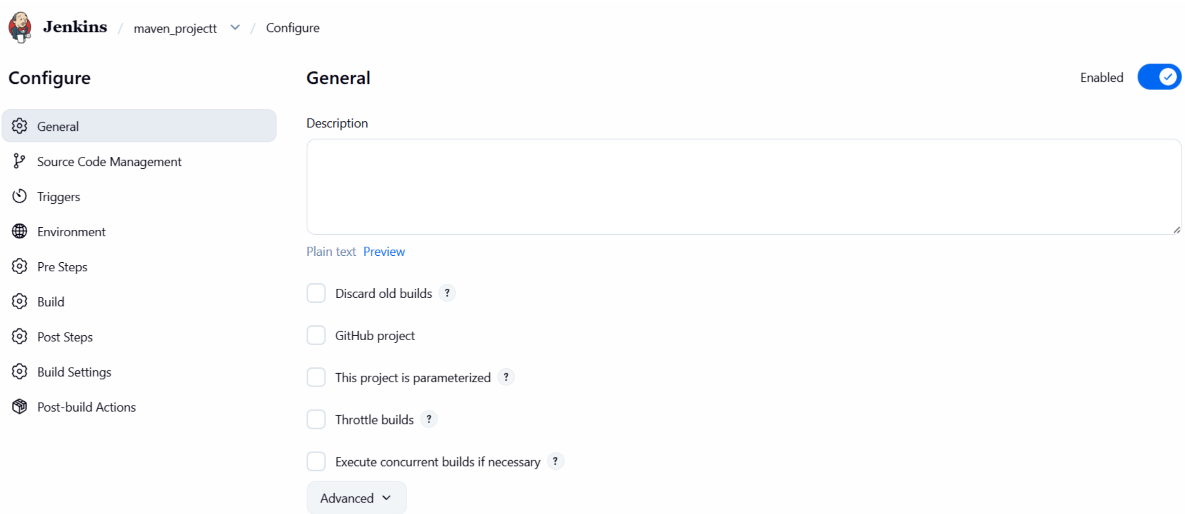

JDK installations
Add JDK
JDK
Name
java-21
JAVA_HOME
/usr/lib/jvm/java-21-openjdk
Install automatically
?

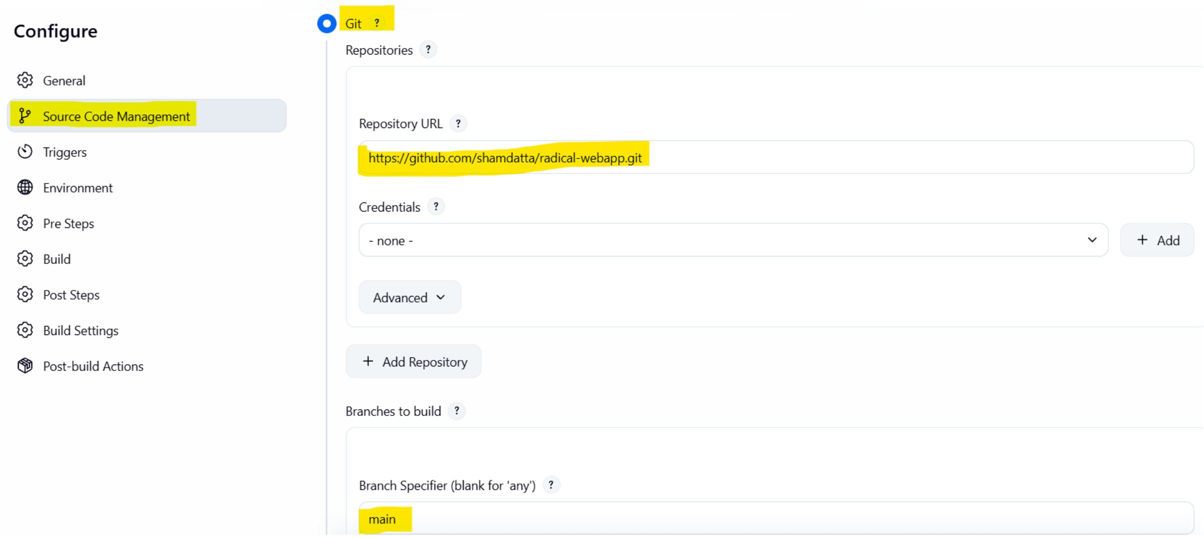
Create a new Jenkins job named maven_project and select Maven Project as the project type.

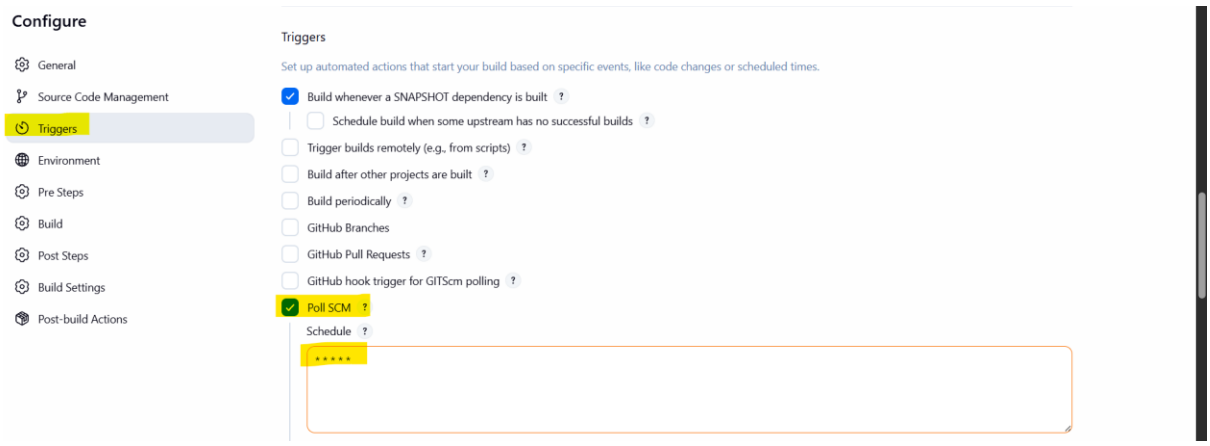


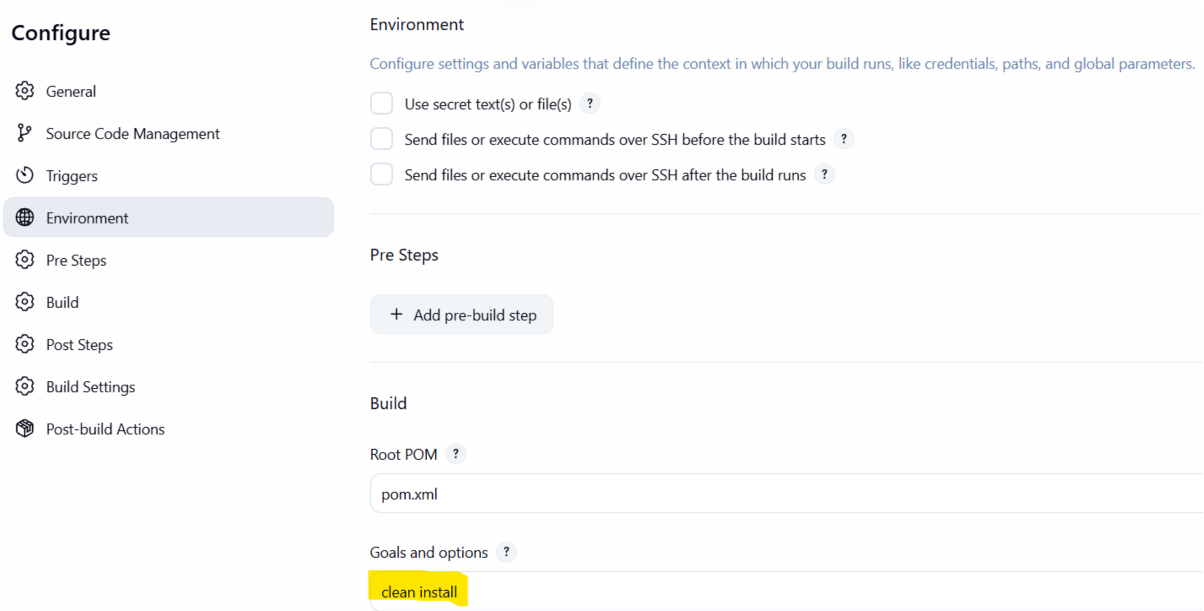
Configure the remaining options as shown in the screenshots below.

Jenkins Job Configuration




Configure
General
Source Code Management
Triggers
Environment
Pre Steps
Build
Post Steps
Build Settings
Post-build Actions
Git ?
Repositories ?
Repository URL ?
https://github.com/shamdatta/radical-webapp.git
Credentials ?
- none -
+ Add
Advanced
+ Add Repository
Branches to build ?
Branch Specifier (blank for 'any') ?
main


Configure
General
Source Code Management
Triggers
Environment
Pre Steps
Build
Post Steps
Build Settings
Post-build Actions
Triggers
Set up automated actions that start your build based on specific events, like code changes or scheduled times.
Build whenever a SNAPSHOT dependency is built ?
Schedule build when some upstream has no successful builds ?
Trigger builds remotely (e.g., from scripts) ?
Build after other projects are built ?
Build periodically ?
GitHub Branches
GitHub Pull Requests ?
GitHub hook trigger for GITScm polling ?
Poll SCM ?
Schedule ?
* * * * *


Configure
General
Source Code Management
Triggers
Environment
Pre Steps
Build
Post Steps
Build Settings
Post-build Actions
Environment
Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.
Use secret text(s) or file(s) ?
Send files or execute commands over SSH before the build starts ?
Send files or execute commands over SSH after the build runs ?
Pre Steps
+ Add pre-build step
Build
Root POM ?
pom.xml
Goals and options ?
clean install

```
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.jar (196 kB at 1.9 MB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18.jar (157 kB at 1.9 MB/s)
[INFO] Installing /var/lib/jenkins/workspace/maven_projectt/pom.xml to /var/lib/jenkins/.m2/repository/in/radical/webapp/radical-webapp/1.0-SNAPSHOT/radical-webapp-1.0-SNAPSHOT.pom
[INFO] Installing /var/lib/jenkins/workspace/maven_projectt/target/radical-webapp.war to /var/lib/jenkins/.m2/repository/in/radical/webapp/radical-webapp/1.0-SNAPSHOT/radical-webapp-1.0-SNAPSHOT.war
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  17.789 s
[INFO] Finished at: 2026-07-07T23:43:36+05:30
[INFO] ------------------------------------------------------------------------
Waiting for Jenkins to finish collecting data
[JENKINS] Archiving /var/lib/jenkins/workspace/maven_projectt/pom.xml to in.radical.webapp/radical-webapp/1.0-SNAPSHOT/radical-webapp-1.0-SNAPSHOT.pom
[JENKINS] Archiving /var/lib/jenkins/workspace/maven_projectt/target/radical-webapp.war to in.radical.webapp/radical-webapp/1.0-SNAPSHOT/radical-webapp-1.0-SNAPSHOT.war
channel stopped
Finished: SUCCESS
```

The Maven project has now been configured successfully as you can see the console output as success above.

Step 12- Tomcat Server Configuration

Configure Hostname

#hostnamectl set-hostname tomcat.example.com

```
[root@tomcatsrv1 ~]# hostnamectl set-hostname tomcat.example.com
[root@tomcatsrv1 ~]#
```

```
[root@tomcatsrv1 ~]# ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.220.136  netmask 255.255.255.0  broadcast 192.168.220.255
        inet6 fe80::20c:29ff:fec7:4e76  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:c7:4e:76  txqueuelen 1000  (Ethernet)
        RX packets 175155  bytes 246984807 (235.5 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 28168  bytes 1831892 (1.7 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 22  bytes 2340 (2.2 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 22  bytes 2340 (2.2 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Install Java

#yum install java-21-openjdk -y

```
Total                                                                                  3.7 MB/s |  50 MB     00:13
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Running scriptlet: java-21-openjdk-headless-1:21.0.11.0.10-2.el9.x86_64                                            1/1
  Preparing        :                                                                                                 1/1
  Installing       : java-21-openjdk-headless-1:21.0.11.0.10-2.el9.x86_64                                            1/2
  Running scriptlet: java-21-openjdk-headless-1:21.0.11.0.10-2.el9.x86_64                                            1/2
  Installing       : java-21-openjdk-1:21.0.11.0.10-2.el9.x86_64                                                     2/2
  Running scriptlet: java-21-openjdk-1:21.0.11.0.10-2.el9.x86_64                                                     2/2
  Running scriptlet: java-21-openjdk-headless-1:21.0.11.0.10-2.el9.x86_64                                            2/2
  Running scriptlet: java-21-openjdk-1:21.0.11.0.10-2.el9.x86_64                                                     2/2
  Verifying        : java-21-openjdk-1:21.0.11.0.10-2.el9.x86_64                                                     1/2
  Verifying        : java-21-openjdk-headless-1:21.0.11.0.10-2.el9.x86_64                                            2/2
Installed products updated.

Installed:
  java-21-openjdk-1:21.0.11.0.10-2.el9.x86_64              java-21-openjdk-headless-1:21.0.11.0.10-2.el9.x86_64

Complete!
[root@tomcatsrv1 ~]#
```

Download Apache Tomcat

#cd /opt

#wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.55/bin/apache-tomcat-10.1.55.tar.gz

#tar -xvzf apache-tomcat-10.1.55.tar.gz

#mv apache-tomcat-10.1.55 tomcat

```
[root@tomcatsrv1 opt]# wget https://archive.apache.org/dist/tomcat/tomcat-10/v10.1.55/bin/apache-tomcat-10.1.55.tar.gz
--2026-07-07 23:58:55--  https://archive.apache.org/dist/tomcat/tomcat-10/v10.1.55/bin/apache-tomcat-10.1.55.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14343328 (14M) [application/x-gzip]
Saving to: ‘apache-tomcat-10.1.55.tar.gz.1’

apache-tomcat-10.1.55.tar.gz.1  100%[====================================================>]  13.68M   749KB/s    in 19s

2026-07-07 23:59:15 (726 KB/s) - ‘apache-tomcat-10.1.55.tar.gz.1’ saved [14343328/14343328]

[root@tomcatsrv1 opt]# tar -xvzf apache-tomcat-10.1.55.tar.gz
apache-tomcat-10.1.55/conf/
apache-tomcat-10.1.55/conf/catalina.policy
apache-tomcat-10.1.55/conf/catalina.properties
apache-tomcat-10.1.55/conf/context.xml
apache-tomcat-10.1.55/conf/jaspic-providers.xml
apache-tomcat-10.1.55/conf/jaspic-providers.xsd
apache-tomcat-10.1.55/conf/logging.properties
apache-tomcat-10.1.55/conf/server.xml
```

```
[root@tomcatsrv1 opt]#
[root@tomcatsrv1 opt]# mv apache-tomcat-10.1.55 tomcat
[root@tomcatsrv1 opt]# ls
apache-tomcat-10.1.55.tar.gz  apache-tomcat-10.1.55.tar.gz.1  tomcat
```

Create Tomcat User… For security reasons, Tomcat should run as a dedicated non-root user.

#groupadd --system tomcat
#useradd -r -g tomcat -d /usr/share/tomcat -s /bin/false tomcat

```
[root@tomcatsrv1 opt]# groupadd --system tomcat
groupadd: group 'tomcat' already exists
[root@tomcatsrv1 opt]# useradd -r -g tomcat -d /usr/share/tomcat -s /bin/false tomcat
useradd: user 'tomcat' already exists
```

Assign Ownership

#chown -R tomcat:tomcat /opt/tomcat

```
[root@tomcatsrv1 opt]# chown -R tomcat:tomcat /opt/tomcat
[root@tomcatsrv1 opt]# ll
total 28020
-rw-r--r--.  1 root   root   14343328 May  5 19:09 apache-tomcat-10.1.55.tar.gz
-rw-r--r--.  1 root   root   14343328 May  5 19:09 apache-tomcat-10.1.55.tar.gz.1
drwxr-xr-x. 10 tomcat tomcat     4096 Jul  8 00:01 tomcat
```

Start Tomcat

cd /opt/tomcat/bin
./startup.sh

```
[root@tomcatsrv1 opt]# cd /opt/tomcat/bin
[root@tomcatsrv1 bin]# ./startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@tomcatsrv1 bin]#
```

Open Firewall

#firewall-cmd --permanent --add-port=8083/tcp
#firewall-cmd --reload

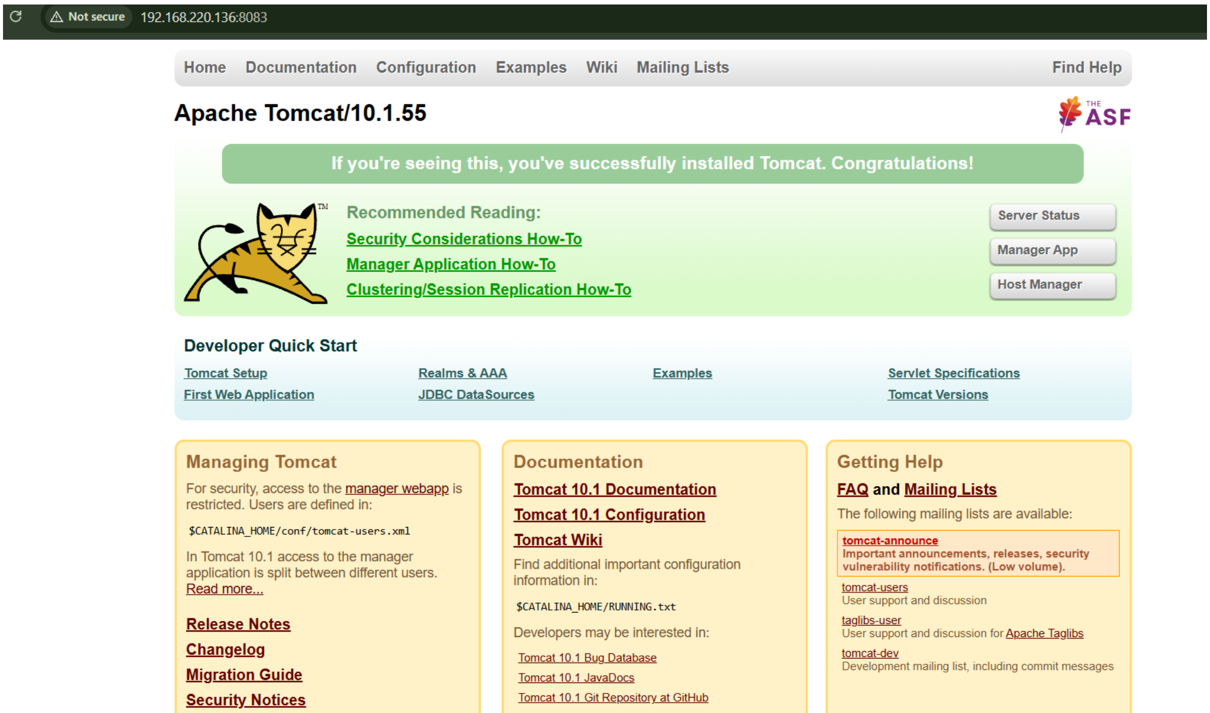
```
[root@tomcatsrv1 bin]# firewall-cmd --permanent --add-port=8083/tcp
success
[root@tomcatsrv1 bin]# firewall-cmd --reload
success
[root@tomcatsrv1 bin]# netstat -antlpu | grep java
tcp6       0      0 :::8083                 :::*                    LISTEN      34210/java
tcp6       0      0 127.0.0.1:8005          :::*                    LISTEN      34210/java
tcp6       0      0 192.168.220.136:8083    192.168.220.1:60208     ESTABLISHED 34210/java
tcp6       0      0 192.168.220.136:8083    192.168.220.1:51731     ESTABLISHED 34210/java
tcp6       0      0 192.168.220.136:8083    192.168.220.1:55124     ESTABLISHED 34210/java
```

Verify

#ss -tulnp | grep 8080

Access from browser

http://<Tomcat-IP>:8080



<u>Configure Users</u>

Edit

/opt/tomcat/conf/tomcat-users.xml

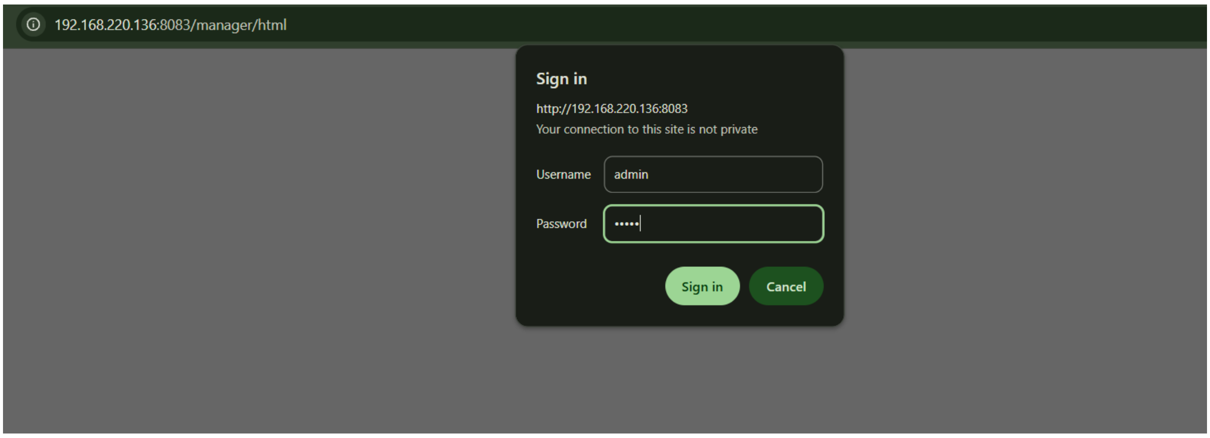
Add

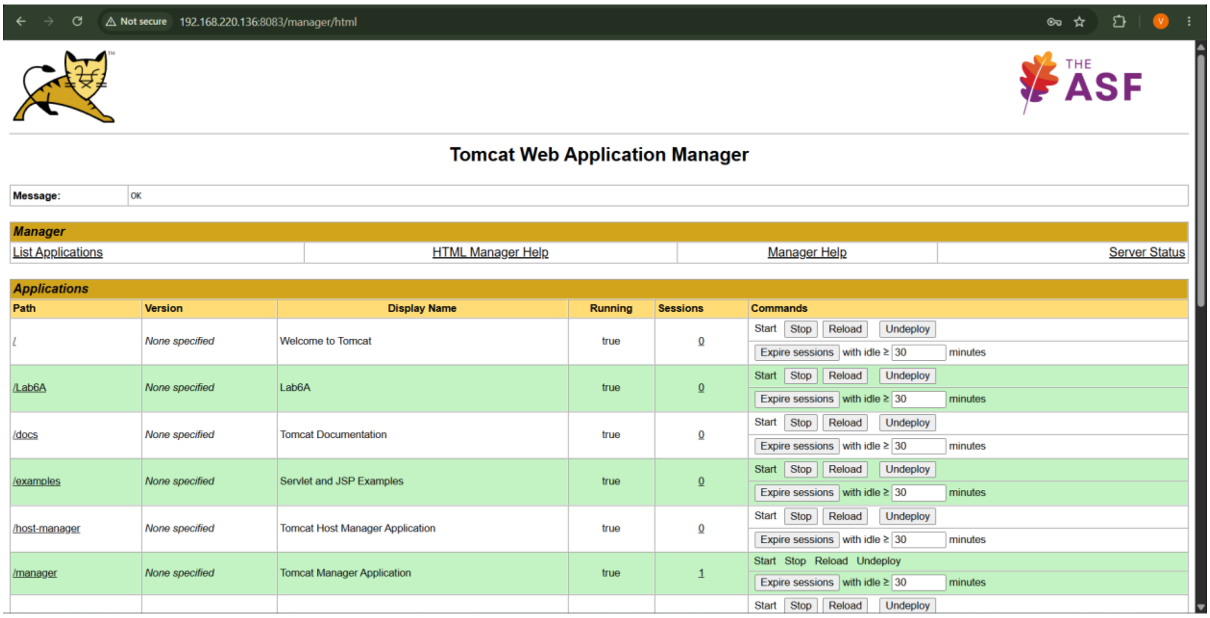
```
<role rolename="manager-gui"/>
<role rolename="manager-script"/>
<role rolename="manager-jmx"/>
<role rolename="manager-status"/>

<user username="admin"
    password="admin"
    roles="manager-gui,manager-script,manager-jmx,manager-status"/>
```

Restart Tomcat.

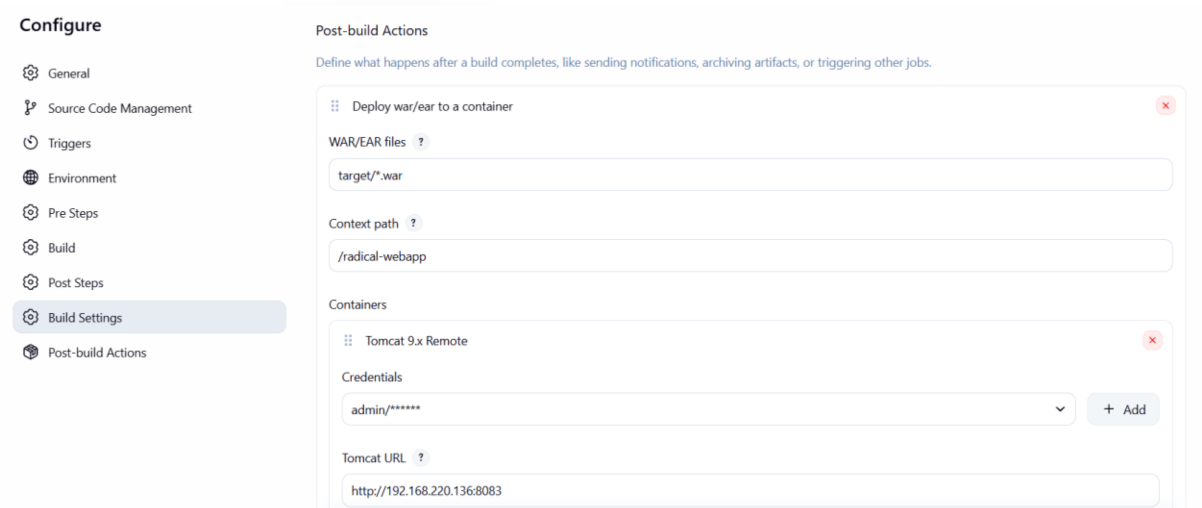
```
[root@tomcat bin]# vi /opt/tomcat/conf/tomcat-users.xml
[root@tomcat bin]#
[root@tomcat bin]#
[root@tomcat bin]# ./shutdown.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
[root@tomcat bin]# ./startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@tomcat bin]#
```

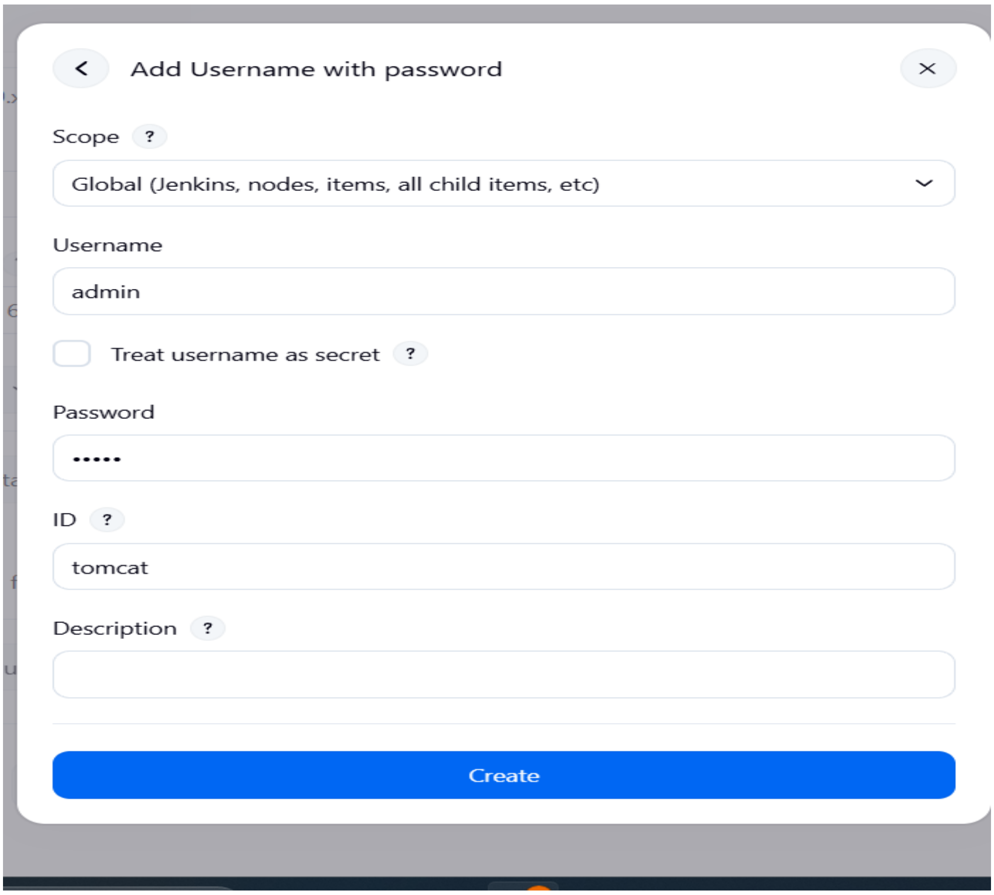


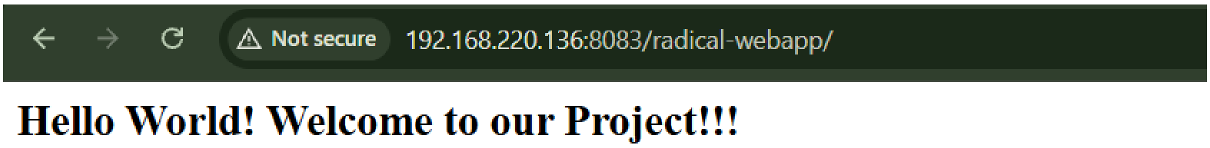


Now for tomcat do these changes in our same 'maven_project'

Deployment Configuration







Conclusion

The CI/CD pipeline has been configured successfully.

The pipeline now performs the following workflow:

1. Developer pushes code to GitHub.
2. Jenkins detects the changes.
3. Jenkins builds the Maven project.
4. A WAR file is generated.
5. The application is deployed to Apache Tomcat.
6. Users can access the deployed application through a web browser.